

Suryansh Sharma

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Bangalore, India

Summary

Final-year B.Tech CSE student at PES University specializing in **AI systems, LLM engineering, and prompt design**. Experienced in building **RAG pipelines, LLM-powered applications, and red-teaming frameworks** for evaluating model robustness and safety. Skilled in designing **end-to-end AI systems with structured outputs, evaluation pipelines, and real-world deployment**.

Education

PES University, Bangalore

B.Tech in Computer Science, CGPA: 8.36

2022 – Present (2026)

Experience

Data Science & ML Intern

June 2025 – August 2025

Naxxatra Sciences and Collaborative Research

- Developed a **CNN-based MRI classification model** for Alzheimer's detection achieving **99% accuracy and recall** with robust preprocessing and evaluation pipelines.

Associate Analyst and GenAI

Feb 2026 – Present

Tech Mahindra (Ongoing)

- Built **GCP-based RAG chatbot workflows** for enterprise AI applications with retrieval and context injection.
- Developed **Chrome extension-based data extraction systems** enabling real-time browser-level AI interactions.
- Conducted **LLM red teaming using PromptFoo and adversarial testing strategies** to evaluate model robustness and security vulnerabilities.

Projects

LLM Red Teaming & Evaluation Platform

AI Safety / LLM Red Teaming

- Engineered an automated LLM security testing platform integrating **Garak** and **PromptFoo** to simulate adversarial attacks (prompt injection, jailbreaks, encoding).
- Developed a **FastAPI backend** to execute configuration-driven workflows and parse complex evaluation runs into structured **JSON vulnerability reports**.

Technologies: Garak, PromptFoo, FastAPI, LiteLLM, Groq, Python

Multimodal RAG Recipe Generator

Generative AI / NLP

- Built a **Retrieval-Augmented Generation (RAG)** pipeline using **FAISS** and **LLaMA** with semantic retrieval to ensure coherent, context-aware recipe generation.
- Extended the architecture to support **multimodal outputs**, generating step-by-step images and cooking videos using diffusion-based models.

Technologies: RAG, FAISS, LangChain, Transformers, Python

Multilingual Propaganda Detection Engine

NLP Systems / ML Engineering

- Architected a **Sequential Transfer Learning pipeline** using **XLM-RoBERTa** to enable cross-lingual (English/Hindi) classification of manipulative social media text.
- Mitigated label noise and a severe 5.5:1 class imbalance by engineering a **custom PyTorch Focal Loss** function and a **synthetic data augmentation** pipeline.
- Optimized large-model training on consumer GPUs via **fp16 mixed-precision**, leveraging **LoRA (PEFT)** to prevent catastrophic forgetting during domain shifts.

Technologies: PyTorch, Hugging Face (Transformers, PEFT), XLM-RoBERTa, Data Augmentation

AI-Powered Project Management Tool

AI Systems Architecture / Backend Engineering

- Engineered **Generative AI pipelines** to drive platform intelligence, featuring real-time "Scope Drift" auditing, predictive readiness scoring, and integrated conversational agents.
- Architected a **secure, high-throughput API layer** with robust relational data models to orchestrate dynamic state synchronization across the application.
- Implemented comprehensive backend infrastructure featuring **Role-Based Access Control (RBAC)** and a unified **object management system** for handling multi-modal digital artifacts.

Technologies: Python, FastAPI, Groq API (Llama-3), SQLAlchemy, Agentic Development, Alembic

Skills

Languages: Python, SQL

AI & LLM: Prompt Engineering, RAG, PEFT (LoRA), Red Teaming, Multimodal AI, Domain Adaptation (DAPT)

Frameworks & Libraries: PyTorch, Hugging Face, LangChain, LiteLLM, FAISS, SQLite, FastAPI, SQLAlchemy, Alembic

Models & APIs: LLaMA, XLM-RoBERTa, Gemini API, Groq

Cloud & Tools: GCP, Docker, Git, CI/CD, GPU Optimization

Core CS: Data Structures, OS, DBMS, Computer Networks